```
#question 1
import pandas as pd
fixed_entries = [
    {
        "question": "what is the annual fee",
        "answer": "The annual fee is Rs 500.",
        "keywords": "fee, charge, annual",
        "category": "billing"
    },
    {
        "question": "how to reset password",
        "answer": "Go to Settings > Reset Password.",
        "keywords": "password, login",
        "category": "account"
    },
    {
        "question": "what are your working hours",
        "answer": "We are open 9 AM to 5 PM.",
        "keywords": "hours, timing, open",
        "category": "general"
    },
    {
        "question": "how can i pay the fee",
        "answer": "You can pay via UPI, card, or net banking.",
        "keywords": "pay, payment, UPI",
        "category": "billing"
    }
]

#roll no = 1024170368
#6%3=0(billing), 8%3=2(general)

my_entries = [
    {
        "question": "how do i update my payment details",
        "answer": "You can update your payment details from the account settings.",
        "keywords": "payment, update, details",
        "category": "billing"
    },
    {
        "question": "where can i find the college help desk",
        "answer": "You can find help desk information on the college portal.",
        "keywords": "help, desk, college",
        "category": "general"
    }
]
all_entries = fixed_entries + my_entries

df = pd.DataFrame(all_entries)
print(df)
```

```
                                question  \
0                 what is the annual fee
1                  how to reset password
2            what are your working hours
3                  how can i pay the fee
4     how do i update my payment details
5  where can i find the college help desk

                                          answer  \
0                        The annual fee is Rs 500.
1                  Go to Settings > Reset Password.
2                        We are open 9 AM to 5 PM.
3        You can pay via UPI, card, or net banking.
4  You can update your payment details from the a...
5  You can find help desk information on the coll...

                  keywords category
0      fee, charge, annual  billing
1          password, login  account
2      hours, timing, open  general
3        pay, payment, UPI  billing
4  payment, update, details  billing
5      help, desk, college  general
```

```
#question 2

def score_query(query):
    query = query.lower()
```

```
    results = []

    for index, row in df.iterrows():

        keywords = row["keywords"].lower().split(", ")

        score = 0

        for keyword in keywords:
            if keyword in query:
                score += 1

        result = row.to_dict()
        result["score"] = score

        results.append(result)

    results.sort(key=lambda x: x["score"], reverse=True)

    return results
```

```
query = "how can i pay my fee"

results = score_query(query)

for result in results:
    print("Question:", result["question"])
    print("Score:", result["score"])
    print()
```

```
Question: what is the annual fee
Score: 1

Question: how can i pay the fee
Score: 1

Question: how to reset password
Score: 0

Question: what are your working hours
Score: 0

Question: how do i update my payment details
Score: 0

Question: where can i find the college help desk
Score: 0
```

```
#question 3
#my personalized entry is "billing"
def same_category(category_name, df):
    return df[df["category"] == category_name]
result = same_category("billing", df)

print(result)
```

```
                            question  \
0              what is the annual fee
3               how can i pay the fee
4  how do i update my payment details

                                          answer  \
0                         The annual fee is Rs 500.
3        You can pay via UPI, card, or net banking.
4  You can update your payment details from the a...

                  keywords category
0       fee, charge, annual  billing
3         pay, payment, UPI  billing
4  payment, update, details  billing
```

```
#question 4

# Pick entry 4
entry_index = 4

# Ask the user for a new keyword
new_keyword = input("Enter a new keyword: ")

# Add the new keyword
df.loc[entry_index, "keywords"] = df.loc[entry_index, "keywords"] + ", " + new_keyword
```

```
# Display the updated DataFrame
print(df)

# Save as CSV
df.to_csv("1024170368_faq_data.csv", index=False)

print("DataFrame saved")
```

```
Enter a new keyword: billing
                               question  \
0                  what is the annual fee
1                   how to reset password
2             what are your working hours
3                   how can i pay the fee
4      how do i update my payment details
5  where can i find the college help desk

                                        answer  \
0                        The annual fee is Rs 500.
1                    Go to Settings > Reset Password.
2                           We are open 9 AM to 5 PM.
3         You can pay via UPI, card, or net banking.
4  You can update your payment details from the a...
5  You can find help desk information on the coll...

                          keywords category
0               fee, charge, annual  billing
1                   password, login  account
2               hours, timing, open  general
3                 pay, payment, UPI  billing
4  payment, update, details, billing  billing
5               help, desk, college  general
DataFrame saved
```

```
#question 5

category_counts = df.groupby("category").size()

print(category_counts)
```

```
category
account    1
billing    3
general    2
dtype: int64
```

```
#question 6

def score_query(query):
    query = query.lower()

    results = []

    for index, row in df.iterrows():

        keywords = row["keywords"].lower().split(", ")

        score = 0

        for keyword in keywords:
            if keyword in query:
                score += 1

        result = row.to_dict()
        result["score"] = score

        results.append(result)

    results.sort(key=lambda x: x["score"], reverse=True)

    highest_score = results[0]["score"]

    if highest_score == 0:
        print("No matching FAQ found.")
        return []

    top_results = []

    for result in results:
        if result["score"] == highest_score:
            top_results.append(result)

    return top_results
```

```
results = score_query("fee")

for result in results:
    print("Question:", result["question"])
    print("Answer:", result["answer"])
    print("Score:", result["score"])
    print()
```

```
Question: what is the annual fee
Answer: The annual fee is Rs 500.
Score: 1
```

```
Start coding or generate with AI.
```